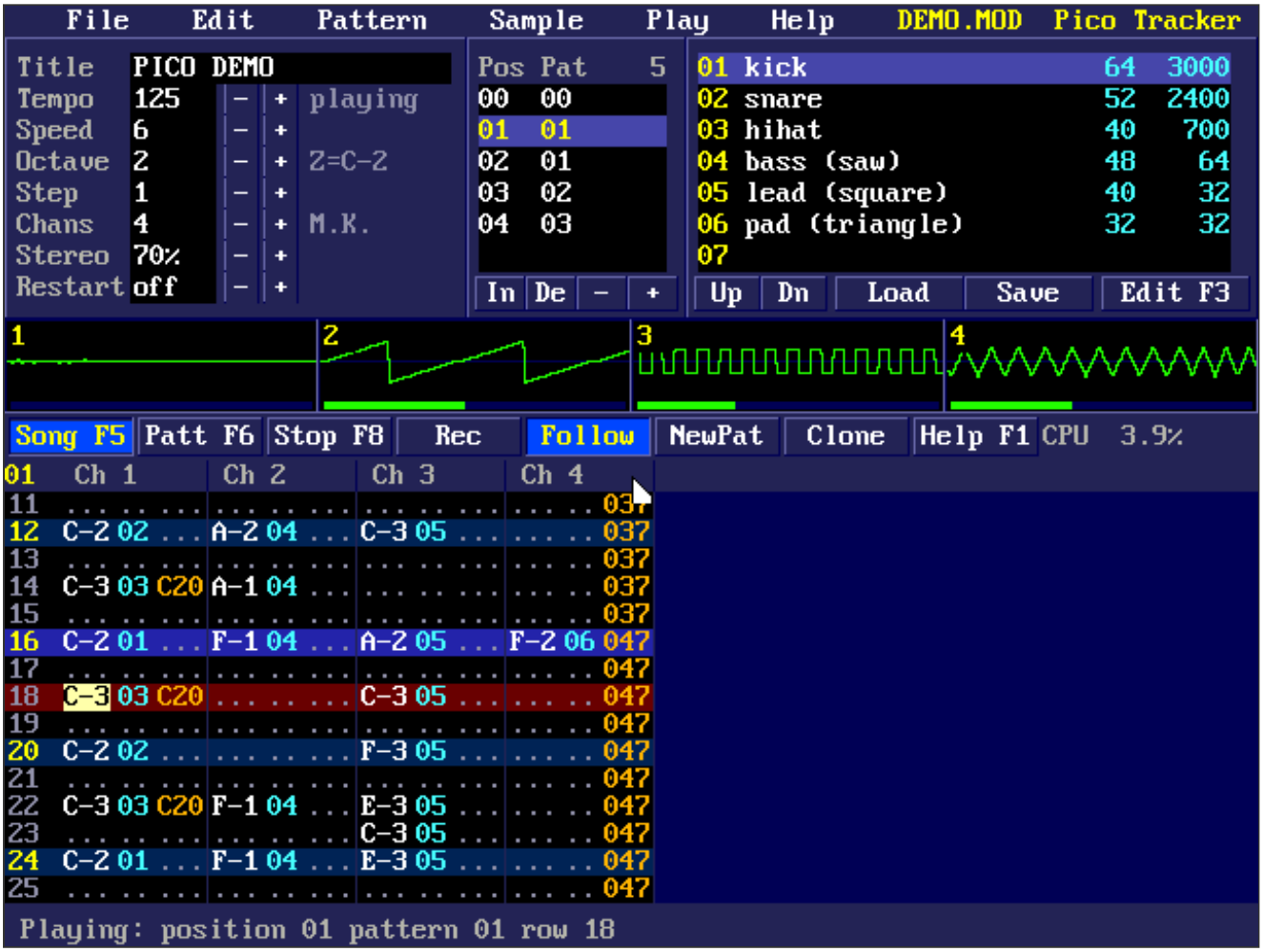
Pico 2 mini OS MOD Tracker

Pico 2 mini OS · October 27, 2026

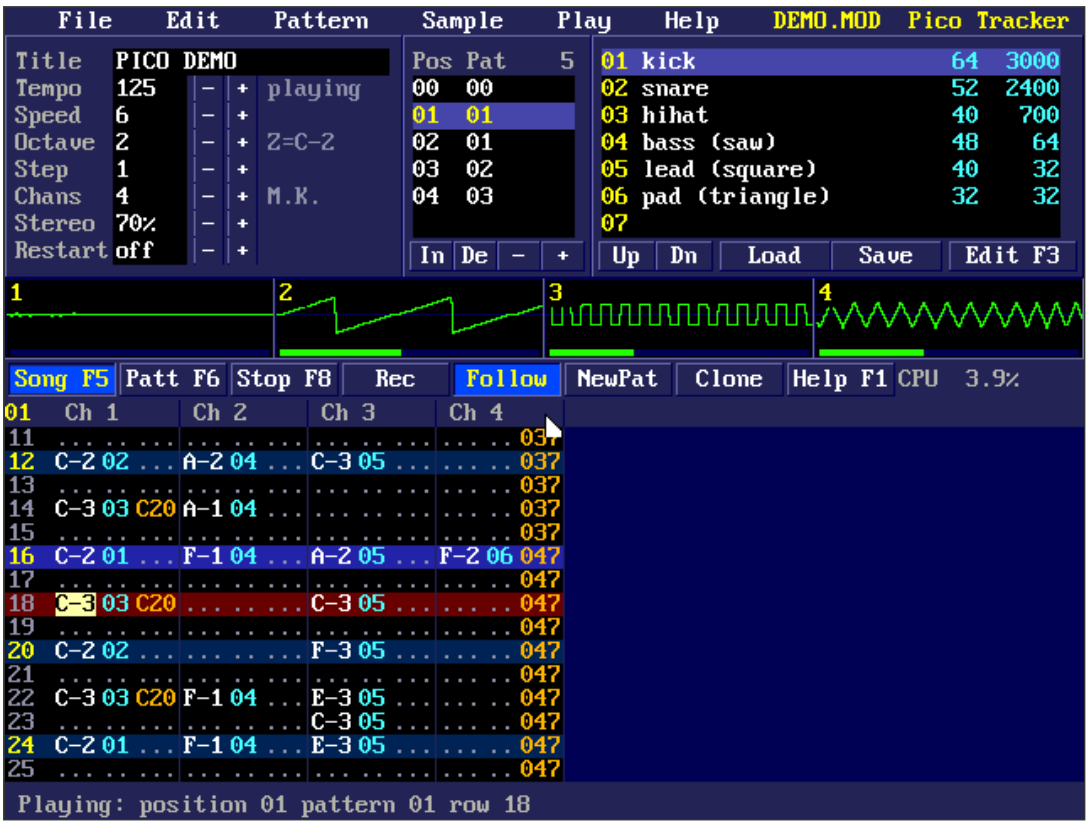


Contents

The tracker
Playing songs: modplay

`tracker [file]` (or `mod`) is a ProTracker style editor for MOD songs (4, 6 or 8 channels), which also loads, edits and saves FastTracker 2 XM songs. `modplay file` plays a MOD or XM song without the editor.

The tracker



`tracker` (or `mod`) edits and plays ProTracker `.MOD` songs on the 640 x 480 graphics screen: `tracker /MUSIC/SONG.MOD` opens one, `tracker` alone starts a new song. `/EXAMPLES/DEMO.MOD` is a short demo song made for this OS. `modplay file.mod` just plays one.

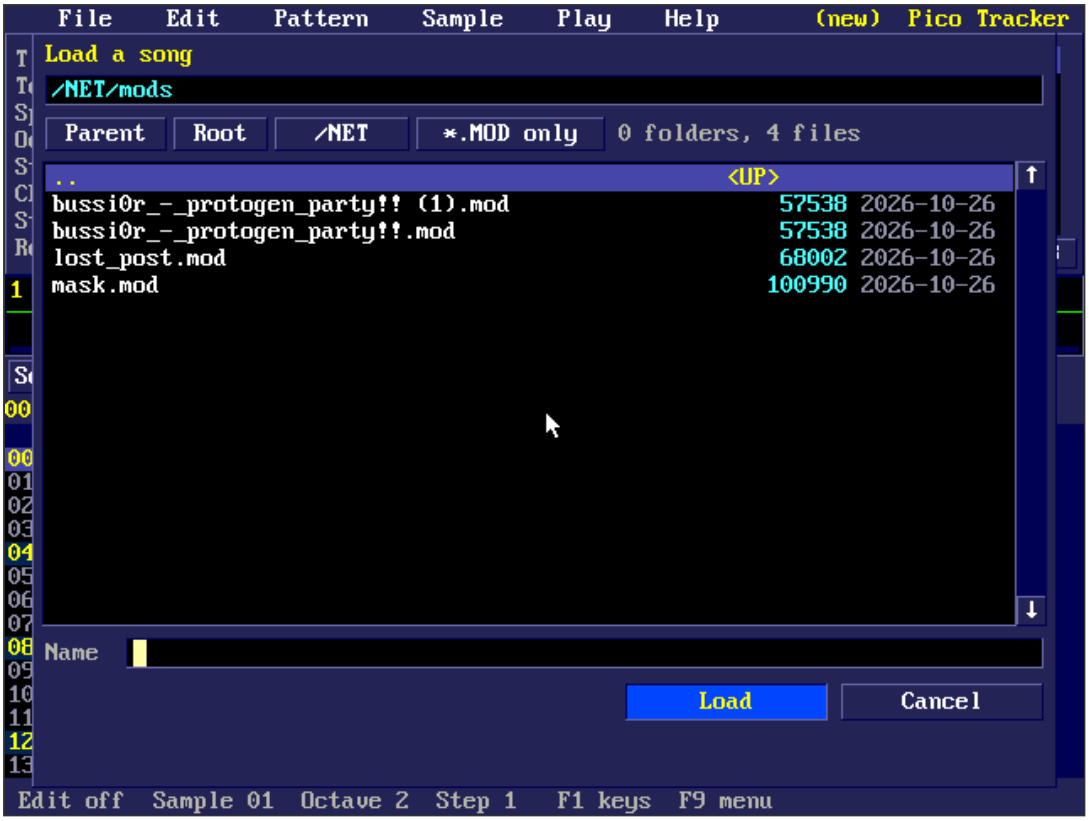
- **Songs:** 4, 6 or 8 channels (M.K., M!K!, FLT4, 4CHN, 6CHN, 8CHN, FLT8, OCTA), 31 samples, and old 15-sample Soundtracker files. Saving gives back the same bytes as were loaded (a 15-sample song is saved in the 31-sample layout). Up to 100 patterns and 128 positions. Songs load and save on the drive and on network shares (see the OS manual), through a file picker.
- **Playback** runs on core 1 at 48 kHz into the HDMI sound, as ProTracker plays: every effect 0-F and E0-EE (EF, invert loop, is ignored), with linear interpolation, Amiga panning (left, right, right, left) and a stereo separation setting. The song keeps playing while you edit.
- **Pattern screen (F2):** song settings (title, tempo, speed, octave, edit step, channels, stereo, restart), the order list, the sample list, a scope and level meter for each channel, the transport buttons, and the pattern (note, sample, effect for each channel). The row playing is red; with Follow on, the view moves with the song.
- **Sample screen (F3):** the sample's name, volume, finetune and loop, its waveform with the loop (orange S and E marks) and a selection, and tools: load (WAV 8/16/24/32-bit or float, stereo mixed to mono; IFF 8SVX; or raw 8-bit), save as WAV (8-bit, with the loop), synthesize (sine, square, saw, triangle, noise), normalize, amplify, reverse, fade in/out, remove DC, clear, crop, cut, silence, loop the selection, halve or double the rate,

copy to another sample, rename. A MOD sample holds at most 131070 bytes: longer files are cut.

- **Undo:** Ctrl-Z undoes and Ctrl-Y redoes, many steps back (up to 64 steps, 3 MB of copies): notes and blocks in the pattern, the order list and song settings, sample and instrument settings, envelopes, the note map and the sample tools. Also in the Edit menu, greyed when there is nothing.
- **Menus:** F9 (or a click on the menu bar): File, Edit, Pattern, Sample (Instrument for XM songs), Play, Help, drawn in the tracker's own style: a raised bar, drop-downs with a shadow, shortcuts on the right, ticks on settings that are on, greyed items that can't be used now, and an underlined letter for each item (press it to choose it). The mouse highlights what it is over; press on a title and let go on an item, or click; Left/Right go to the next menu. The **right button** opens the Edit menu (pattern screen) or the Sample/Instrument tools (F3) where you click. Questions, names and numbers are asked in dialogs of the same style. Menus and dialogs are drawn off screen and put up in one go, and the mouse arrow is drawn by the video output itself, so nothing flickers as the mouse moves over them.

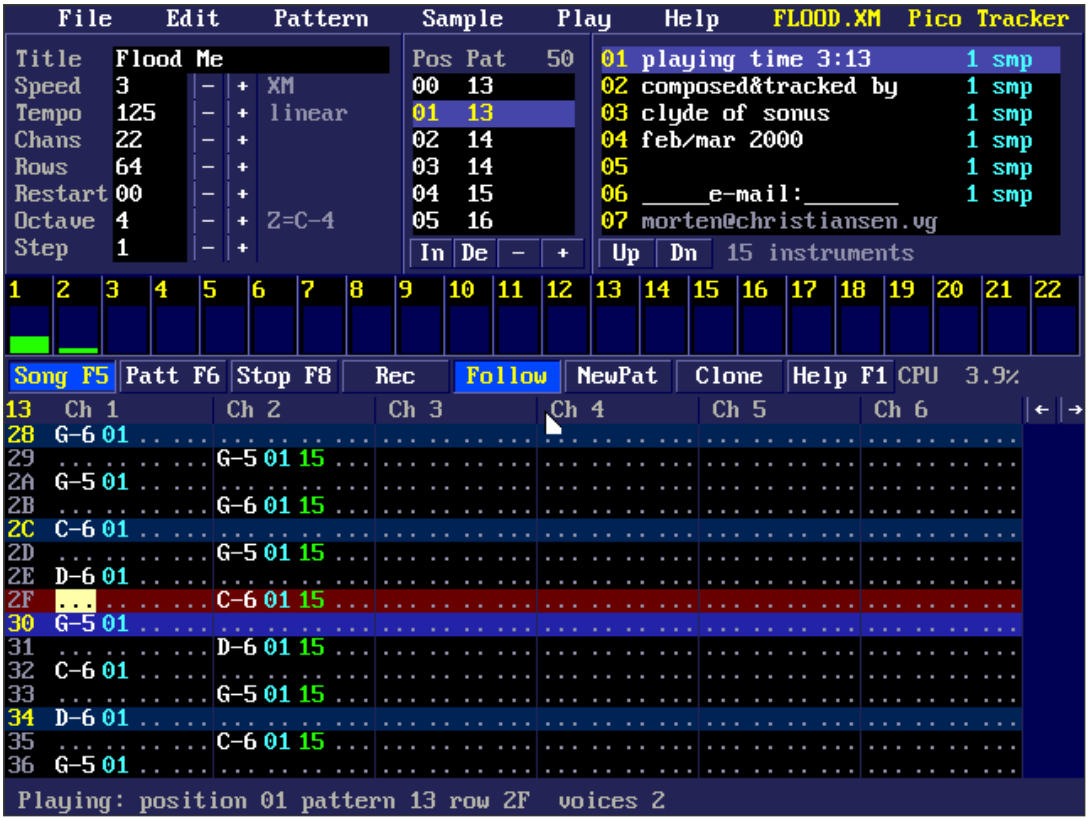


- **Loading and saving** (songs and samples) use a disk panel in the tracker's style: the folder, its folders and files with sizes and dates, a scroll bar, Parent / Root / each mounted share (`/NET`), a filter (`*.MOD` or all files), a name field and Load/Save. Arrows, PgUp/PgDn, Home/End move; Enter opens a folder or takes the file (or the name typed, which may be a path like `/NET/mods`); Backspace with the name empty goes up; Tab switches the filter; Esc cancels. With the mouse: click, double-click, the wheel, the scroll bar's arrows, track and thumb. Saving over a file asks first.



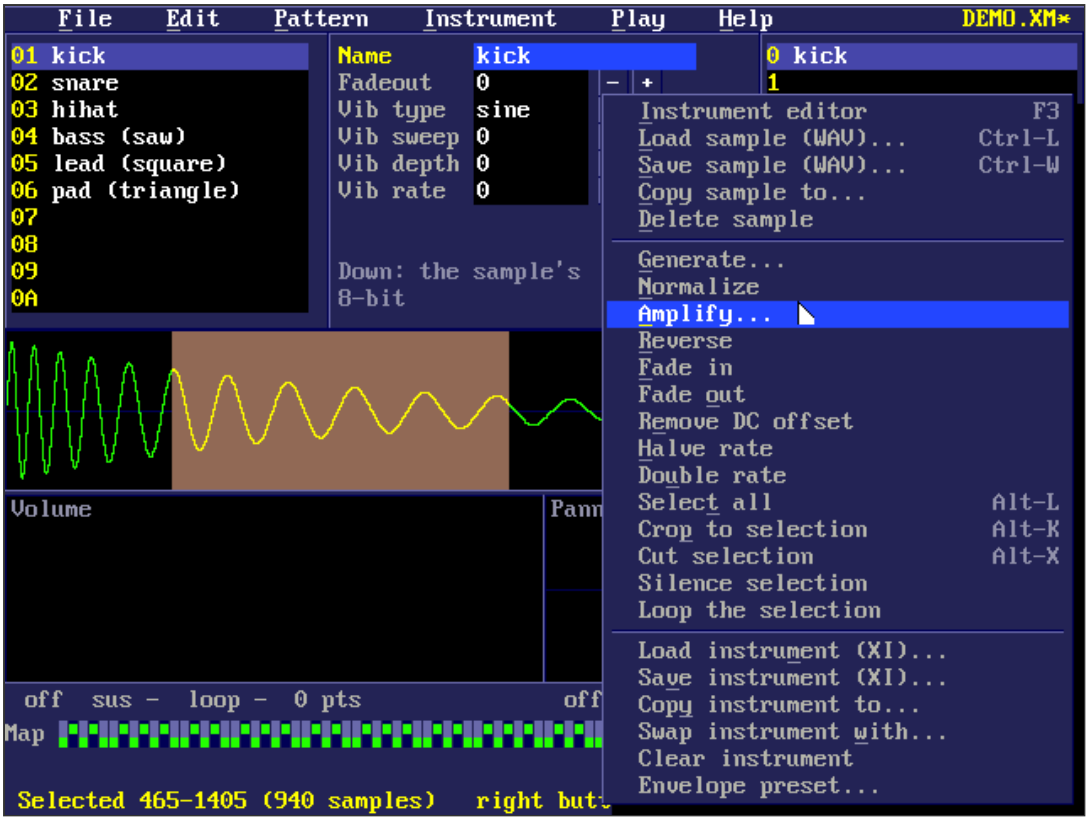
XM songs (FastTracker 2) load into the tracker too, and Ctrl-N offers a new XM song (8 channels). The screen is the same, with XM's fields: note, instrument, volume column and effect in each channel (six channels show at once; the view follows the cursor, and the arrow buttons by the channel headings scroll it), 2 to 32 channels, patterns of 1 to 256 rows (the Rows setting), and the instrument list. The level meters show every channel (click one to mute it, right-click to solo it). Notes go up to octave 7 (F11 and F12 change the octave); `1` or the key left of it writes a key off (`===`). In the volume column `0`-`4` start a volume (00-40) and `-` `+` `d` `u` `s` `v` `p` `l` `r` `m` its commands (slide down/up, fine down/up, vibrato speed/depth, pan, pan slide left/right, tone portamento); effects are typed as FT2 shows them (0-9, A-Z). Blocks, transpose, rows, new and cloned patterns, play, follow and record work as for MODs. Saving writes an XM (a file loaded and saved unchanged comes back byte for byte). F3 is the instrument screen for XM songs.

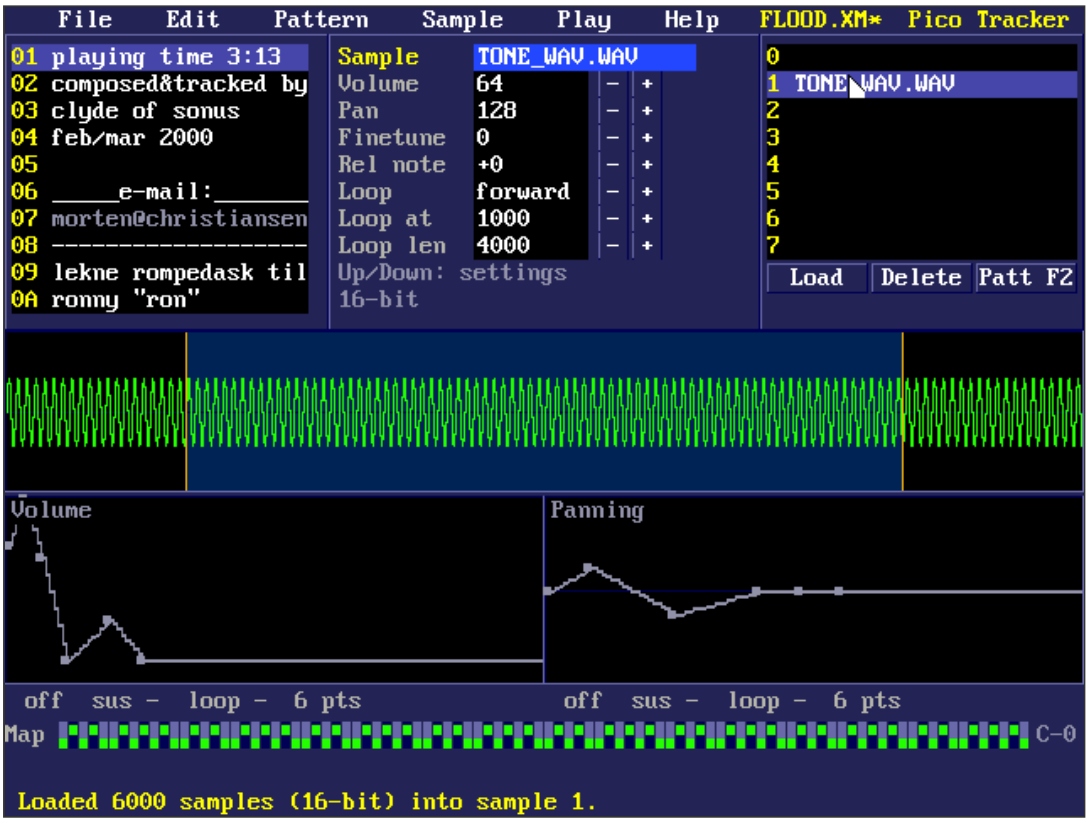
MOD to XM: File > Convert to XM, or Save As with a `.XM` name, makes the MOD into an XM that plays the same (checked against libopenmpt): Amiga frequencies, ProTracker's C-1..B-3 as FT2's C-3..B-5, one instrument per sample with its sound and loop, finetunes in FT2's steps (and the small difference between the Amiga's and FT2's base pitch), arpeggios in FT2's order, the Amiga's left-right-right-left panning (at the Stereo setting) as panning commands in the volume column, and the other effects as they were.



The XM instrument screen (F3; F2 goes back): the instruments on the left (`-` and `=` choose; one past the last makes a new one), the chosen instrument's settings (name, fadeout, auto-vibrato type, sweep, depth, rate) and its samples (up to 16; `[` and `]` choose), the chosen sample's settings (name, volume, pan, finetune, relative note, loop off/forward/ping-pong, loop start and length) and its waveform with the loop (drag its orange ends). Up/Down go through the settings, Left/Right (PgUp/PgDn: 16 times as far) change them, Enter types them. **Load** (or Ctrl-L) reads a WAV (8 to 32-bit or float, stereo mixed) into the chosen sample as 16-bit, with its loop; **Delete** removes a sample. The volume and panning **envelopes**: click an empty place to add a point, drag points, right-click one to delete it (12 at most); their on/off, sustain point and loop (from-to) are settings too (Left/Right step through the points). The **note map** below them is a 96-key piano coloured by sample: click or drag across it to give those notes the chosen sample. The piano keys play the instrument.

Drag across the waveform to **select** part of it (Alt-L all, Alt-U none). The **Instrument menu** (or the right button on this screen) has the sample tools, on 16-bit sound: generate (sine, square, saw, triangle, noise, in tune at C-4), normalize, amplify, reverse, fade in/out, remove DC, halve or double the rate (the relative note follows, so it sounds the same), select all, crop (Alt-K), cut (Alt-X), silence, loop the selection, copy the sample into another instrument, delete it; **save the sample** as a 16-bit WAV at the rate C-4 plays it, with its loop (and loading a WAV sets the relative note and finetune from its rate). For the instrument: load and save **.XI** files (FT2's instrument files, with all their samples), copy it to another number, swap it with another, clear it, and **envelope presets** (piano, organ, pad, pluck, tremolo; pan sweep, autopan). Everything here can be undone.





- Leaving with unsaved changes asks whether to save them. F1 shows all the keys.

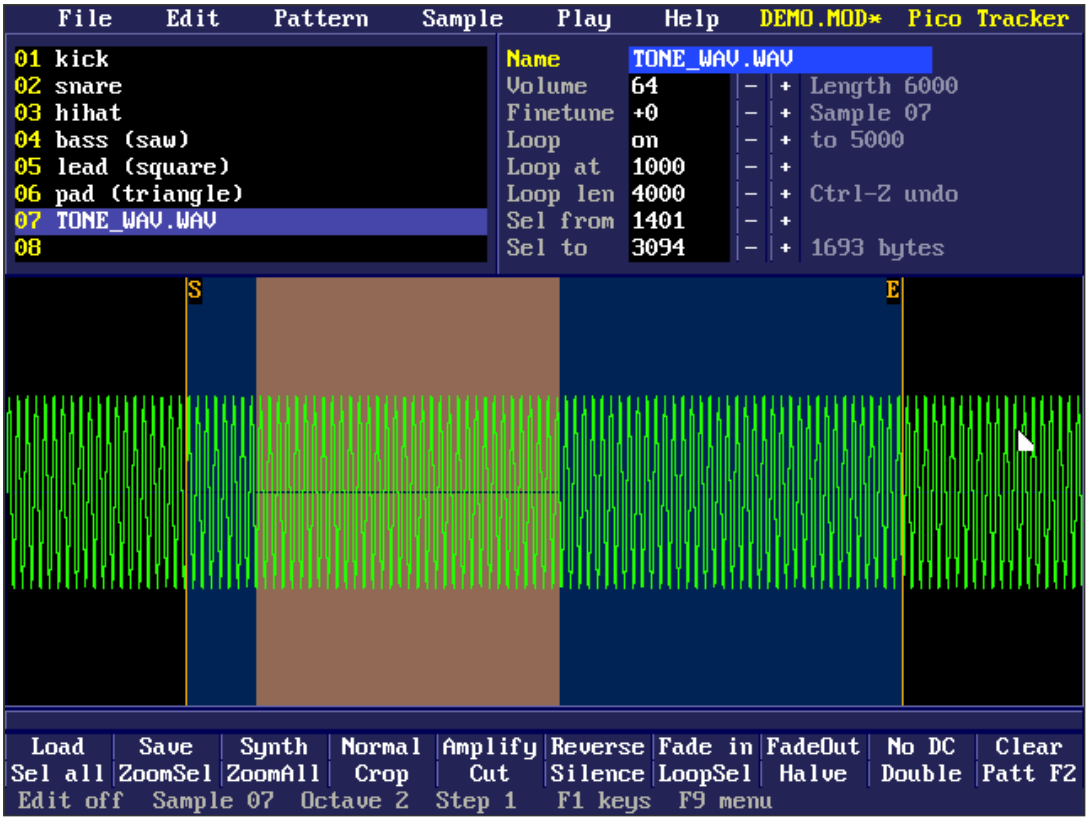
Key	Action
F1 / F2 / F3	Help / pattern screen / sample screen
F4	Keyboard to the pattern, the order list, or the song settings
F5 / F6 / F7 / F8	Play the song / loop the pattern / play from the cursor / stop

Key	Action
Space	Stop if playing, otherwise record (edit) mode on or off
Z S X D C V G B H N J M , L . ; /	Piano keys: the base octave (C D E F G A B C D E)
Q 2 W 3 E R 5 T 6 Y 7 U I 9 O 0 P	Piano keys, an octave up
F11 / F12, `[` / `]`, `-` / `=`	Octave down/up, edit step down/up, sample down/up
`<` / `>`	Previous / next song position
Arrows, Tab, PgUp/PgDn, Home/End	Move (Tab: next channel; 16 rows; first/last row)
0-9, A-F	In the sample, effect and parameter columns: type the digit (record on)
Del, Ins, Backspace, Enter	Clear the field; insert or delete a row in the channel; play the row
Alt-B, Alt-E, Alt-L, Alt-U	Begin / end block, mark channel (twice: pattern), unmark
Alt-C, Alt-X, Alt-V, Alt-M	Copy, cut, paste, mix paste (only fills empty fields)
Alt-Q / Alt-A, Alt-W / Alt-S	Transpose a semitone up/down, an octave up/down (block or cursor)
Alt-I / Alt-D	Insert / delete a row in all channels
Alt-N / Alt-K / Alt-R	New pattern, clone the pattern, clear the pattern
Alt-F / Alt-G	Write the speed / tempo as an Fxx effect at the cursor
Ctrl-S / Ctrl-O / Ctrl-N	Save, load, new
Ctrl-T / Ctrl-L / Ctrl-U	Mute the channel, solo it, unmute all
Ctrl-F	Follow the song on or off
Ctrl-Q, Esc	Quit

On the sample screen Up/Down choose a setting, Left/Right (PgUp/PgDn: 16 times as far) change it and Enter types it; the piano keys play the sample; Alt-L selects all, Alt-U unselects, Alt-Z zooms to the selection, Alt-O zooms out, Alt-+ / Alt-- zoom, Alt-, / Alt-. scroll, Alt-X cuts and Alt-K crops; Ctrl-L loads and Ctrl-W saves a sample.

With a mouse: click the buttons, fields (and their - and + buttons), samples, positions and pattern cells; drag in the pattern to mark a block; drag on the waveform to select, or drag its S and E marks to move the loop; drag the bar under it to scroll. The wheel moves the pattern, changes the value under the pointer, or zooms the waveform. Click a scope or a channel heading to mute it; right-click a scope to solo it; right-click elsewhere for the menus. Double-click a sample to edit it, or a pattern cell to switch record mode.

The status bar shows the mixer's share of core 1 ("CPU", about 4% for four channels) and `!` if the sound ever skipped.


File Edit Pattern Sample Play Help DEMO.MOD* Pico Tracker
01 kick
02 snare
03 hihat
04 bass (saw)
05 lead (square)
06 pad (triangle)
07 TONE_WAV.WAV
08
Name TONE_WAV.WAV
Volume 64 - + Length 6000
Finetune +0 - + Sample 07
Loop on - + to 5000
Loop at 1000 - +
Loop len 4000 - + Ctrl-Z undo
Sel from 1401 - +
Sel to 3094 - + 1693 bytes
S
E
Load Save Synth Normal Amplify Reverse Fade in FadeOut No DC Clear
Sel all ZoomSel ZoomAll Crop Cut Silence LoopSel Halve Double Patt F2
Edit off Sample 07 Octave 2 Step 1 F1 keys F9 menu

Playing songs: modplay

`modplay file` plays a MOD or XM (FastTracker 2) song from the drive or a share once through the HDMI sound; any key stops it. BASIC's `MUSIC` statement uses the same player.